



Properties, crystallization mechanism and microstructure of lithium disilicate glass–ceramic

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ARTICLE INFO

Article history:

Received 20 May 2009

Received in revised form 24 October 2009

Available online 4 January 2010

Keywords:

Bioglass

Glass–ceramics

Nucleation

ABSTRACT

In this study, lithium disilicate glass–ceramic in the $\text{TiO}_2\text{--ZrO}_2\text{--Li}_2\text{O--CaO--Al}_2\text{O}_3\text{--SiO}_2$ system was investigated for dentistry applications by incorporation of P_2O_5 and Nb_2O_5 as nucleation agent. The influence of the particles size (nano and submicron size) and nucleating agents on the crystalline phases, microstructure, crystallization mechanism and mechanical properties were investigated. Our data indicated that in ceramic glass with nano and submicron P_2O_5 , the main crystalline phase was lithium disilicate. The results also showed that change of P_2O_5 particle's size had significant effect on the crystalline phases and microstructure. By replacement of submicron P_2O_5 with submicron Nb_2O_5 , crystallization mechanism was changed from volume to surface crystallization. Due to the nature of crystalline phases as well as the directional crystallization and coarsening, many cracks were observed in the samples containing submicron Nb_2O_5 . The composition of base glass was changed and nano Nb_2O_5 was added. As a result of these changes, the crystallization mechanism was altered.

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1. Introduction

Glass–ceramics can be produced by molten glass and converting it into a fine grained ceramic by an appropriate heat treatment [1]. The well known, wide-range properties of glass–ceramics can be modified in a predictable way by changing the composition and a suitable heat treatment [1,2]. The final crystalline phases depend on the composition of glass, specie of nucleating agent and the heating methods. In other words, the properties of glass–ceramics are determined by the crystallization of phases precipitated from the glasses and their microstructures, which depend on the composition of the parent glass as well as thermal treatment and addition of nucleating agents [3–6]. When developing a new kind of glass–ceramics, it is very important to control the crystallization of the glass. Regarding the kinetics, when glassy phases transform to crystalline phase(s), it is necessary to overcome a certain energy barrier, which is called the activation energy of crystallization (E). E is one of the important kinetics parameters to judge crystallization capability of any glasses. If this energy were lower, it would be easier for the glasses to crystallize. Similarly, the crystal growth index (n) can also be applied to describe well the capability of crystallization of any glasses. The higher the crystal growth index (n), the easier the crystallization. When the index

(n) is less than three, the surface crystallization mechanism will be dominated; when n is more than three, it is expected to have bulk crystallization mechanism [7–9,18,19]. Despite the huge efforts on the application of metallic materials in dentistry, since 1990s, efforts to develop biomaterials for restorative dentistry have been concentrated on producing metal-free systems [10]. An important milestone in this respect was reached in the development of glass–ceramics containing lithium disilicate ($\text{Li}_2\text{Si}_2\text{O}_5$). Lithium disilicate glass–ceramic was the first glass–ceramic that Stockey [11] developed. This glass–ceramic is particularly suitable for fabrication of anterior dental bridges [12,13]. A common nucleating agent for this type of glass–ceramics is P_2O_5 , which promotes heterogeneous nucleation and produces a fine grained interlocking morphology after heat treatment [3,14–17].

The present study is focused on the nucleation stage in lithium disilicate glass–ceramic and the effects of size and specie of nucleating agents on the crystallization mechanism, microstructure and properties of the this system.

2. Experimental procedure

The glasses were prepared according to the compositions summarized in Table 1. Laboratory reagent-grade Li_2CO_3 , SiO_2 , Al_2O_3 , ZrO_2 , TiO_2 and CaCO_3 powders were chosen as the raw materials. P_2O_5 and Nb_2O_5 were also added as nucleation agents with nano and submicron sizes. The particle size distribution of Nb_2O_5 and

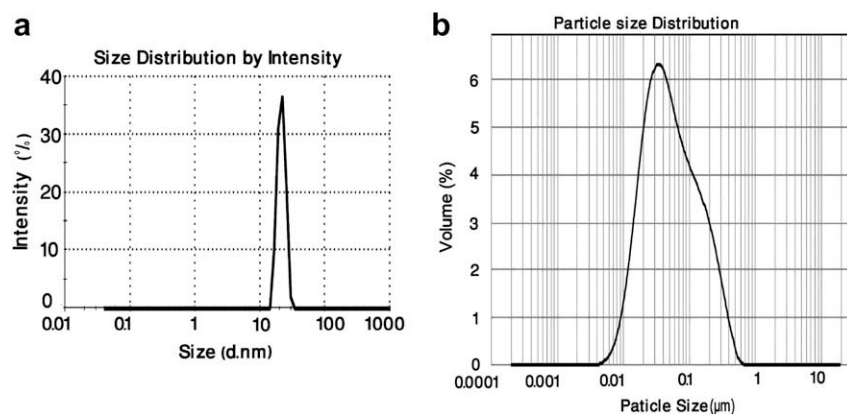
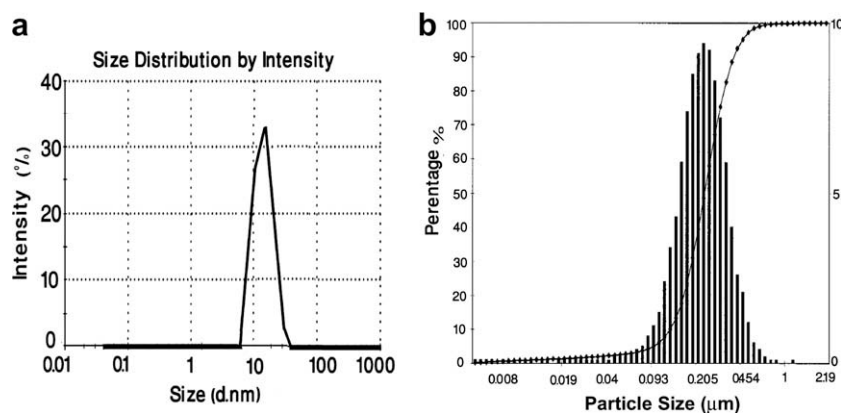
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Table 1

The compositions of GPn, GP, GN and GNn glasses (mol%).

Glass	TiO ₂	SiO ₂	ZrO ₂	Al ₂ O ₃	Submicron P ₂ O ₅	Nano P ₂ O ₅	CaO	Li ₂ O	Nano Nb ₂ O ₅	Submicron Nb ₂ O ₅
GPn	1.5	65.5	2.3	2.8	–	5.3	6.2	16.4	–	–
GP	1.5	65.5	2.3	2.8	5.3	–	6.2	16.4	–	–
GN	1.5	65.5	2.3	2.8	–	–	6.2	16.4	–	5.3
GNn	–	69.3	–	4.8	–	–	6.4	17.3	2.2	–

**Fig. 1.** Particle size distribution of P₂O₅ powder: (a) nano P₂O₅; and (b) submicron P₂O₅.**Fig. 2.** Particle size distribution of Nb₂O₅ powder: (a) nano Nb₂O₅; and (b) submicron Nb₂O₅.

P₂O₅ powders was measured using a particle size analyzer (Master-sizer 2000, Malvern, UK).

The particle size distribution of the nucleation agents are presented in Figs. 1 and 2. The samples were prepared as following. The batches of samples were melted in a high-alumina crucible in an electric furnace at 1400 °C for 2 h in air. The melt was then poured into a mould. The glass specimens were annealed immediately at 450 °C for 1 h to relieve internal stress, and then the furnace was cooled down. The crystallization mechanism, crystalline phases and microstructure were investigated using differential scanning calorimetry (DSC), X-ray diffraction (XRD) and scanning electron microscopy (SEM) techniques. DSC analysis was performed using a calorimeter (Setaram Instrumentation) with Al₂O₃ crucibles in air. The glass powders were heated from room temperature to 1100 °C with rates of 5, 10, 15 and 20 °C min^{−1} in order to detect the nucleation and crystallization temperatures and to evaluate the crystallization mechanism. After heat treatments, the types of the crystalline phases were characterized by XRD (Bruker, D8ADVANCE). The 2θ step increment was 0.04° with an interval time of 2 s per step over the range of 10–70° (λ Cu Kα = 1.5406 Å).

Microstructural and morphological studies of the crystalline phases were conducted using SEM (VEGA/TESCAN) on polished and etched samples with approximately 2 vol.% HF solution for 10 s. For the determination of the flexural strength of the final glass–ceramics, groups of rod specimens (eight rods per group) were tested according to the standard procedure [DIN EN 843-1 (1995)] with a universal testing apparatus (Instron-1186). The rods 2.5 × 2.5 × 25 mm³ were cut from heat-treated glass blocks with a saw were ground with SiC 600 grinding paper. The three-point flexural strength test was conducted with a span length of 20 mm and a crosshead speed of 1.5 mm/min to have the specimen fail after 5–15 s of testing time.

3. Results

3.1. DSC results

The effect of type and size of nucleating agents on the crystallization are presented in Fig. 3. As it is clear from this figure, the

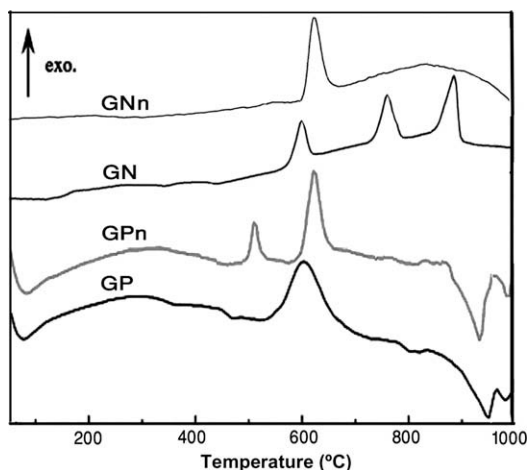


Fig. 3. DSC results of the specimens, heating rate 10 K min⁻¹.

crystallization peak shifted towards higher temperature with decreasing the size of P₂O₅ particles to nano size. In the DSC curve for the GPn sample, two different peaks are detected at 510 and 625 °C. Besides these two temperature, 780 °C was also chosen for heat treatment. The crystalline phases of GP sample were appeared at 610 °C, but two other temperatures '510 and 780 °C' were also chosen for comparison of the nucleation and crystallization temperatures of GPn with GP. GN sample had three crystallization peaks between 600 and 900 °C, thus this sample was heat-treated in an electric furnace at 610, 770 and 880 °C. The crystallization peak of GNn sample was appeared at 620 °C.

DTA and DCS technologies are widely used thermal methods for studying the isothermal and more commonly, the non-isothermal crystallization kinetics of glasses [7–9]. The Johnson–Mehl–Avrami (JMA) equation can be applied to describe the process of crystal growth in a glass. The equation is:

$$\ln(1 - x) = (kt)^n \quad (1)$$

where x is the volume fraction of crystallized phase at time t , n the Avrami exponent related to the mechanism of crystallization, and k is the reaction rate constant, related to the absolute temperature T , by Arrhenius type equation [7–9]:

$$K = \nu \exp(-E/RT) \quad (2)$$

where ν is the frequency factor, R the gas constant and E is activation energy of crystal growth. Starting with Eqs. (1) and (2), non-isothermal crystallization kinetics of glass can be described by the following expression [18]:

$$\ln[(T_p)^2 \alpha^{-1}] = E/R(T_p)^{-1} + \ln[E R^{-1} \nu^{-1}] \quad (3)$$

where T_p is the crystallization peak maximum temperature in DTA and DSC curves, α is the heating rate of DTA and DSC, R the gas constant and E is the activation energy of crystal growth. Values of E can be calculated by Eq. (3) by plots of $\ln[(T_p)^2 \alpha^{-1}]$ versus $(T_p)^{-1}$ (as shown in Fig 4). The results are given in Table 2. The Avrami parameter n was calculated by the Augis–Bennett equation [19]:

$$n = 2.5 R(T_p)^2 (E \times \Delta T)^{-1} \quad (4)$$

where ΔT is the full width of the exothermic peak at the half maximum intensity. As it was mentioned, the crystal growth index (n) can also be applied to describe well the capability of crystallization of any glasses. It has been shown that the higher the crystal growth index (n), the easier the crystallization. The obtained values for index (n), calculated by Eq. (4) for GP, GPn and GN were 3.02, 3.1 and

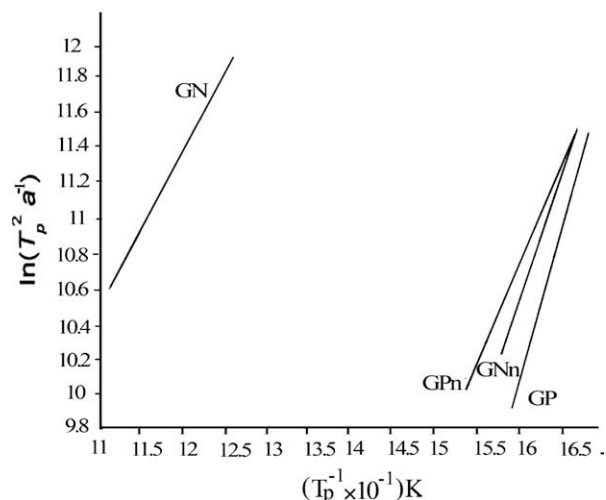


Fig. 4. Plots of $\ln[(T_p)^2 \alpha^{-1}]$ versus $(T_p)^{-1}$ for the glasses.

2.2, respectively (Table 2). These data indicates that the crystallization mechanism was changed from bulk to surface crystallization with changing of specie of nucleating agent. By changing the composition of base glass and by using nano Nb₂O₅ as nucleating agent, crystallization mechanism was then changed from surface to volume crystallization. The values of n calculated for GNn was about 3.3 which mean that the crystallization mechanism was shifted from surface to bulk crystallization with changing of the composition of base glass.

3.2. XRD results

The XRD patterns of the samples are shown in (Figs. 5–9) and the precipitated crystalline phases are summarized in Table 3. Our data indicated that the size of P₂O₅ particles had a significant effect on the kind and volume fraction of crystalline phases of the lithium disilicate glass–ceramics. Under identical experimental conditions, the intensity of the major diffraction peaks of Li₂Si₂O₅ crystals was increased with decreasing the size of P₂O₅ particles in GPn (at 625 °C), which indicated that the relative content of Li₂Si₂O₅ crystals was increased. However, the relative content of LiAl(SiO₃)₂ slightly decreased from GP (at 610 °C) to GPn (at 625 °C). For both GP (at 780 °C) and GPn (at 780 °C), the intensity of the major diffraction peaks of Li₂Si₂O₅ crystals was decreased and the intensity of the major diffraction peaks of LiAlSi₃O₈ crystals was increased. Our observation indicated that Li₂SiO₃ crystals were disappeared in GPn and a new crystalline phase, i.e., Li₃PO₄ crystals were emerged.

According to the XRD patterns of the sample with submicron Nb₂O₅, in the first stage of heat treatment and at 610 °C, three phases of spodumene, wollastonite (CaSiO₃) and LiNb₃O₈ were crystallized. It seems to us that the major crystalline phase in this stage was spodumene. In these samples and at 770 and 880 °C, crystalline phases consisted of spodumene and wollastonite, while spodumene was again the main crystalline phase.

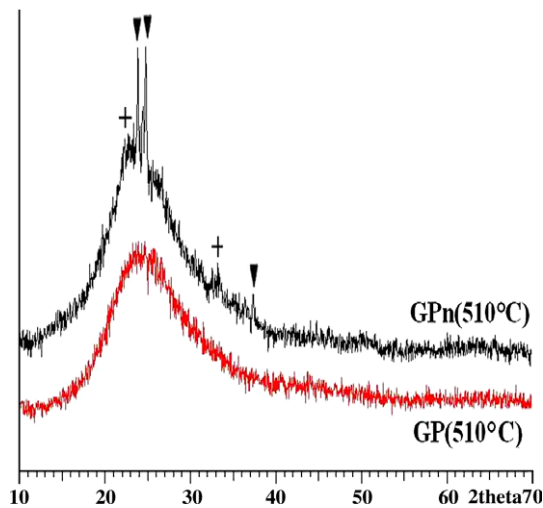
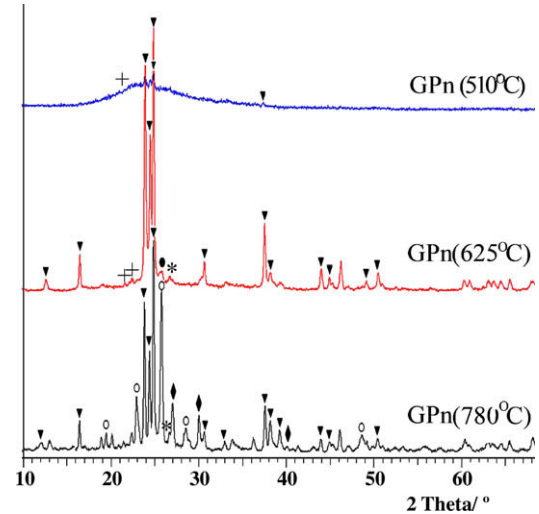
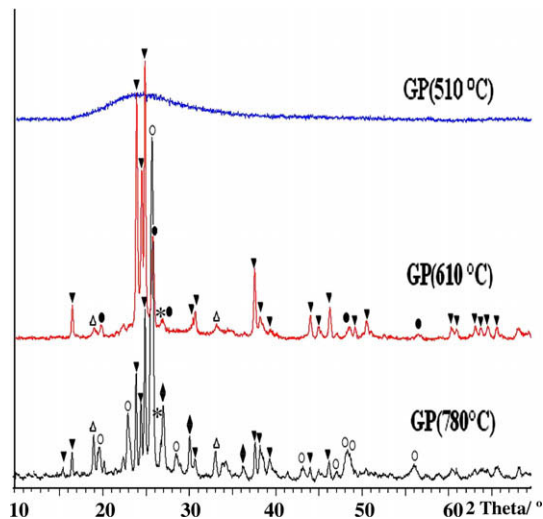
In the sample with nano Nb₂O₅, crystalline phases were determined to be spodumene and lithium metasilicate. According to XRD results, by changing of the nucleating agent and the composition of base glass, the crystalline phases have been changed.

3.3. Morphology of the system

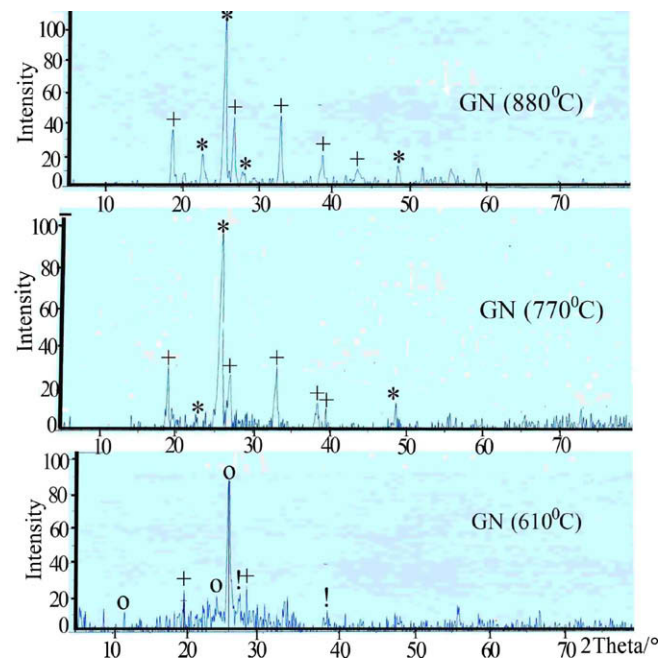
In the next step the morphology of the system was studied and results are presented Fig. 10(f) and (g). The samples with nano and submicron P₂O₅ showed volume crystallization after heat

Table 2 T_p (K), E and index n values of the all samples.

Sample	5 (k min ⁻¹)	10 (k min ⁻¹)	15 (k min ⁻¹)	20 (k min ⁻¹)	E (KJ mol ⁻¹)	n
GP	595	610	620	628	278	3.02
GPn	600	625	640	651	220	3.1
GN	860	880	895	905	300	2.2
GNn	600	620	632	639	210	3.3

**Fig. 5.** Comparison of XRD patterns of the GP (at 510 °C) and GPn (at 510 °C) samples: (+) Li_3PO_4 ; (▼) $\text{Li}_2\text{Si}_2\text{O}_5$.**Fig. 7.** XRD patterns of the sample with nano P_2O_5 : (♦) CaSiO_3 ; (○) $\text{LiAlSi}_3\text{O}_8$; (●) $\text{LiAl}(\text{SiO}_3)_2$; (+) Li_3PO_4 ; (▼) $\text{Li}_2\text{Si}_2\text{O}_5$; (*) SiO_2 .**Fig. 6.** XRD patterns of the sample with submicron P_2O_5 : (♦) CaSiO_3 ; (○) $\text{LiAlSi}_3\text{O}_8$; (●) $\text{LiAl}(\text{SiO}_3)_2$; (▲) Li_2SiO_3 ; (▼) $\text{Li}_2\text{Si}_2\text{O}_5$; (*) SiO_2 .

treatment. The morphology of the precipitated crystals was needle-shaped in GP sample at 610 °C. Crystal size was decreased with decreasing size of P_2O_5 particles, and the morphology of $\text{Li}_2\text{Si}_2\text{O}_5$ crystals was also changed from needle-like to spherical-like in GPn sample at 625 °C. According to SEM results of GP and GPn samples at 780 °C, in addition to disilicate lithium, flat-shaped crystals of spodumene were also observed. The surface crystallization was observed in the samples with submicron Nb_2O_5 as nucleating agent. In these samples, crystals grew from the surface of samples radially into the volume. So it is obvious that the specie of nucleating agent has had a significant effect on the microstructure

**Fig. 8.** XRD patterns of the sample with submicron P_2O_5 : (+) CaSiO_3 ; (○) $\text{LiAl}(\text{SiO}_3)_2$; (*) $\text{LiAlSi}_3\text{O}_8$; (!) LiNb_3O_8 .

and crystallization mechanism of the lithium disilicate glass-ceramic. With utilizing submicron Nb_2O_5 as nucleating agent, the crystallization mechanism was shifted from volume to surface crystallization. In fact, in this step, a monolithic surface crystallization of the glass was observed. It is shown in Fig. 10(f) and (g) that the crystalline phases have been formed from directional

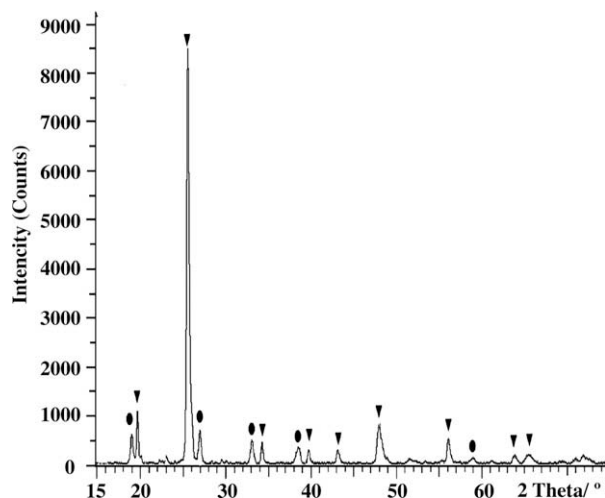


Fig. 9. XRD patterns of the sample with nano P_2O_5 : (●) Li_2SiO_3 ; (▼) $LiAlSi_3O_8$.

Table 3

XRD results of the samples.

Specimen	Crystalline phase
GP (510 °C)	–
GP (610 °C)	$LS_2 + LS + SiO_2 + LiAl(SiO_3)_2$
GP (780 °C)	$LS_2 + LS + SiO_2 + LiAlSi_3O_8 + CaSiO_3$
GPn (510 °C)	$LS_2 + Li_3PO_4$
GPn (625 °C)	$LS_2 + Li_3PO_4 + SiO_2 + LiAl(SiO_3)_2$
GPn (780 °C)	$LS_2 + SiO_2 + LiAlSi_3O_8 + CaSiO_3$
GN (610 °C)	$LiAl(SiO_3)_2 + LiNb_3O_8 + CaSiO_3$
GN (770 °C)	$LiAlSi_3O_8 + CaSiO_3$
GN (880 °C)	$LiAlSi_3O_8 + CaSiO_3$
GNn (620 °C)	$LS + LiAlSi_3O_8$

Crystalline phases: $LS_2 = Li_2Si_2O_5$, ICDD: 40-0376; $LS = Li_2SiO_3$, ICDD: 29-0828; Li_3PO_4 , ICDD: 45-0747; SiO_2 -cristobalite, ICDD: 46-1045; $LiAlSi_3O_8$, ICDD: 35-0794; $CaSiO_3$, ICDD: 43-1460; $LiAl(SiO_3)_2$, ICDD: 31-0706; $LiNb_3O_8$, ICDD: 36-307.

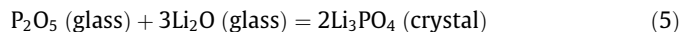
crystallization and coarsening. However due to the crystalline phases directionality and coarsening phenomena, many cracks were observed in the samples. However, in GNn sample, plate-like spodumene crystals were observed.

3.4. Flexural strength

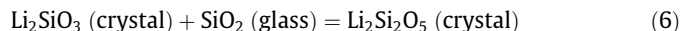
The strengths of the glass-ceramics samples are shown in Table 4. The flexural strength of the samples was increased with decreasing of the particles size of P_2O_5 . Since the samples with submicron Nb_2O_5 showed surface crystallization and cracks formation after heat treatment, they were not suitable for rod preparation and mechanical testing. These results highlight the importance of the nature of nucleating agent and their effects on the mechanical properties. By changing the composition of the base glass and by using nano Nb_2O_5 , suitable samples were prepared. Our data indicated that the strength of GNn sample was decreased with respect to the previous samples.

4. Discussion

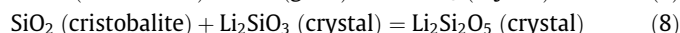
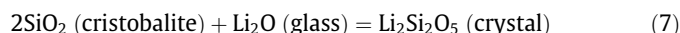
P_2O_5 is an effective nucleating agent in the SiO_2 - Li_2O system. As it has been pointed out in the literature, pure LS_2 glass has no tendency towards amorphous phase separation, but with small additions of P_2O_5 , phase separation will occur [16,17,20–25]. Therefore the Li-rich and Si-rich phases can form and P_2O_5 can react with Li_2O in Li-rich regions to form the Li_3PO_4 crystal nuclei which act as nucleation sites [16] as it has been shown in reaction (5).



It is reasonable to assume that (as it been discussed in a few papers) [16–20] at the beginning, Li_2SiO_3 precipitated on the Li_3PO_4 crystal nuclei in Li-rich regions, and the growth of $Li_2Si_2O_5$ was at the expense of consuming Li_2SiO_3 as shown in reaction (6).



In the GP sample, the amount of Li_3PO_4 crystals, which were made through reaction (5), was quite low. Therefore XRD result of this sample did not indicate the existence of Li_3PO_4 crystals. However the Li_2SiO_3 was precipitated on the small Li_3PO_4 crystal nuclei in Li-rich regions, and the growth of $Li_2Si_2O_5$ was taken place at the expense of consuming Li_2SiO_3 which is shown in reaction (6). In GPn sample, amount of Li_3PO_4 crystal nuclei was high enough to be detected by XRD and also induced cristobalite to precipitate on them. In the GPn sample, the cristobalite crystalline phase not only provided a mechanism of $Li_2Si_2O_5$ (reaction 7) but also consumed the precipitated Li_2SiO_3 crystals to form $Li_2Si_2O_5$ crystals (reaction 8).



According to the XRD results of GP (510 °C) and GPn (510 °C), the nucleation temperature was decreased and the crystallization rates was increased, with decreasing size of P_2O_5 particles to nano scale. At this temperature, the GP sample was remained amorphous but, nucleation was taken placed in GPn sample.

The glass-ceramics with spodumene solid solution as the main crystalline phase exhibit a low coefficient of thermal expansion of $3-7 \times 10^{-7} K^{-1}$ [26–28]. In GPn (at 625 °C) sample, volume fraction of spodumene was less than that of GP (at 610 °C); by which the coefficient of thermal expansion got closer to that of natural teeth. As it is known in the literature, properties of glass-ceramics depend on type, amount, size and distribution of crystalline phase in the base glass. Differences in the coefficient of thermal expansion of different phases can result in tension at the grain boundaries causing micro cracks on surfaces. Non-isotheropic thermal contraction can also cause micro cracking, which initiates at the grain boundaries and then develops [29].

In SiO_2 Al_2O_3 Li_2O system (LAS), the phase type depends on the amount of Al_2O_3 in the composition. If the amount of Al_2O_3 were less, spodumene would be crystallized which has low thermal expansion coefficient and high resistance to thermal shock [30]. In LAS glass-ceramics, the ultra low thermal expansion is due to the main crystalline phases: β -quartz solid solution (Li_2O - Al_2O_3 - $2SiO_2$) and spodumene (Li_2O - Al_2O_3 - $4SiO_2$), but there are some disadvantages: they have low strength (100 MPa for high-quartz and 140 MPa for spodumene) [31].

According to the XRD results of GP (780 °C) and GPn (780 °C) samples, the main phase in GPn (780 °C) was disilicate lithium while in GP (780 °C), the main phase was spodumene. As a result, the third stage of heat treatment which can cause detritions of the properties is not necessary.

The morphology of $Li_2Si_2O_5$ crystals plays an important role in determining the mechanical properties. The appropriate needle-shaped morphology improved the mechanical properties, which is in accordance with the previous results [16]. As Fig. 10 shows the interlocking and needle-shaped $Li_2Si_2O_5$ crystals of GP had grown sufficiently, which was favor to the mechanical properties. As a result, the improved mechanical properties with submicron P_2O_5 as nucleating agent were attributed to the interlocking structure of the elongated $Li_2Si_2O_5$ crystals. These needle-shaped crystals were homogeneously embedded in the glass matrix in an interlocking structure, and they play a major role in hindering crack propagation and elevating the flexural strength. In this microstructure, the growth of micro cracks can be avoided if they

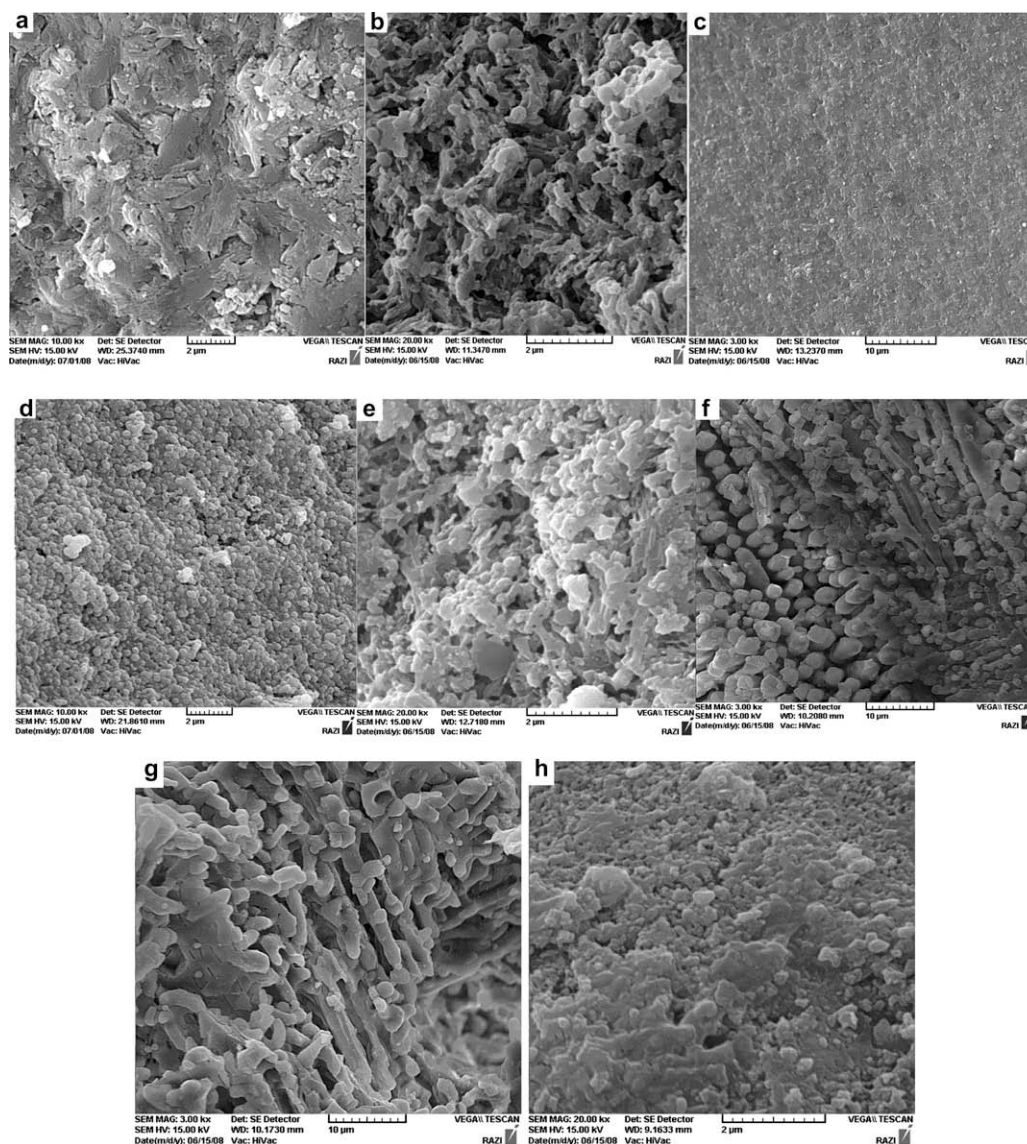


Fig. 10. SEM photographs of surfaces of four samples with nano P_2O_5 and submicron sizes and with nano Nb_2O_5 and submicron sizes: (a) GP (at 510 °C) with 100,000× magnification; (b) GP (at 610 °C) with 100,000× magnification; (c) GP (at 780 °C) with 100,000× magnification; (d) GPn (at 510 °C) with 100,000× magnification; (e) GPn (at 625 °C) with 100,000× magnification; (f) GPn (at 780 °C) with 100,000× magnification; (g) GN (at 780 °C) with 100,000× magnification; (h) GN (at 880 °C) with 100,000× magnification; and (i) GNN (at 620 °C) with 200,000× magnification.

Table 4

Flexural strengths of glass–ceramics samples.

Specimen	Flexural strengths (MPa)
GP (610 °C)	200
GP (780 °C)	170
GPn (625 °C)	280
GPn (780 °C)	225
GN (610 °C)	–
GN (770 °C)	–
GN (880 °C)	–
GNN (620 °C)	153

move vertically to needle crystals. If they move in parallel to needle crystals, it can not be stopped. Therefore growing of micro cracks can be prevented by modifying morphology of crystallized phases from needle-shaped to spherical-shape which results in higher flexural strength.

By changing the kind of nucleating agent to submicron Nb_2O_5 , crystalline phases, microstructure and crystallization mechanism were changed and the major crystalline phase became spodumene. In this sample, surface crystallization was observed.

5. Conclusion

The effects of nucleating agent on the crystallization behavior, mechanical property and microstructure of lithium disilicate glass–ceramics were discussed. The results showed that:

1. In the glass–ceramics with P_2O_5 as nucleating agent, volume crystallization was dominant with $Li_2Si_2O_5$ as major crystalline phase.
2. With decreasing the size of P_2O_5 particles, the relative content of $Li_2Si_2O_5$ crystals was increased and the microstructure of glass–ceramic was changed therefore the lithium disilicate glass–ceramic, (GPn sample) showed better mechanical properties.

3. By using the submicron Nb₂O₅ as nucleating agent, the temperature of heat treatment, major crystalline phase, crystallization mechanism and microstructure were changed.
4. In glass–ceramic with submicron Nb₂O₅ as nucleating agent, crystallization mechanism changed from volume to surface crystallization. Due to the directional crystallization and coarsening, many cracks were observed in this sample.
5. By changing the composition of base glass and by using the nano Nb₂O₅ as nucleating agent, crystallization mechanism was changed from surface to volume crystallization.

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